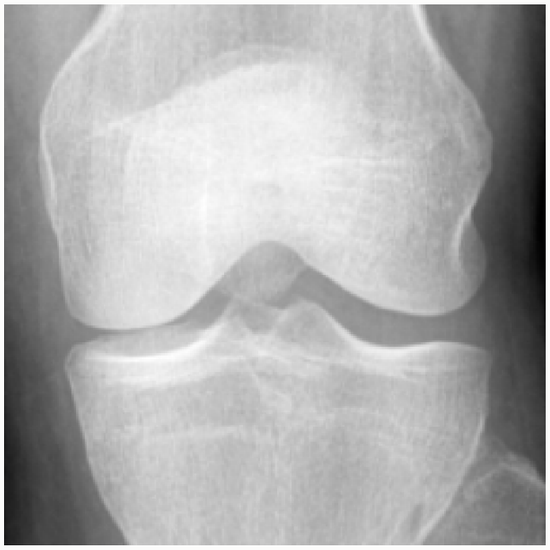
Pipeline της Ομάδας 1 (Χαρακτηριστικά Σχήματος): shape_features()
(OpenCV — cv2.HuMoments, cv2.findContours, cv2.threshold)

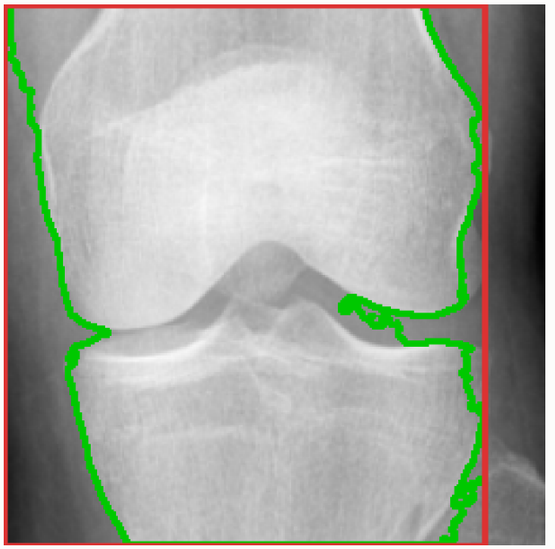
KL0 (υγιές)
Αρχική (grayscale)



Otsu threshold (binary)



Περίγραμμα + bounding box



area=48134 perimeter=1209 aspect_ratio=0.89 extent=0.83
hu1=3.16 hu2=7.39 hu3=9.97 hu4=10.00
hu5=-10.00 hu6=-10.00 hu7=-10.00

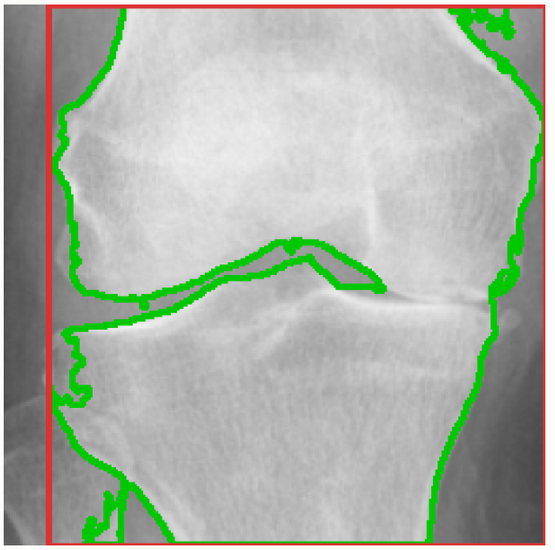
KL4 (σοβαρή παραμόρφωση)
Αρχική (grayscale)



Otsu threshold (binary)



Περίγραμμα + bounding box



area=47272 perimeter=1608 aspect_ratio=0.92 extent=0.79
hu1=3.16 hu2=7.60 hu3=9.96 hu4=10.00
hu5=10.00 hu6=10.00 hu7=-10.00